Hazeshaman

2026-10-14 · 08:00 MYT

MALAYSIA · HAZE RIGHT NOW

UNHEALTHY

Highest API

177

Pasir Gudang

Stations > 100

27 / 68

26 above 150

Stations at closure risk

13 / 68

highest: Pasir Gudang, Johor (83%)

Schools likely affected

2,616

closure chance × schools, state by state · 1,213,805 students

Model estimate from hourly DOE API – confirm with official APIMS and KPM announcements.

Tomorrow morning outlook

The dashboard in one screen



1. **Live badge** – green = the newest hourly DOE report is under 2 h old; amber/red = the feed is lagging or stale.
2. **Haze right now** – the sky, skyline and band name show how bad the air is across Malaysia at this hour.
3. **State tiles** – highest API per state, coloured by band. Click one to open that state.
4. **Live data / What-if** – Live follows the newest report; What-if lets you type your own readings.
5. **State filter** – the whole page follows it: "Malaysia" (all states) or one state. On a phone, use the round button at the bottom-right.
6. **Tabs** – one topic per tab: Overview · Forecast · Schools · Announce time · Data check · Analysis. The haze view stays above as a one-line strip.
7. **National outlook** – tomorrow morning: stations at closure risk and the schools, students and teachers behind them.

STEP 2 · FILTER BY STATE

Malaysia or one state – on desktop and on your phone



Choose a state: the sky, the outlook, the schools and the station list all follow your choice. The station picker then unlocks:



On a phone the same list opens from the round button (bottom-right) as a bottom sheet:



Round button stays in reach while you scroll



Tap a state; the page filters at once

1. **State button** – shows what the page is filtered to. Click it to open the list.
2. **State list** – "Malaysia" first, then every state. Each row shows its highest live API (coloured by band) and Open / Watch / Close status.
3. **Station picker** – after choosing a state, pick one of its stations or leave it on "Highest risk (auto)".
4. **Round button** (phones) – always bottom-right; opens the list.
5. **Bottom sheet** – tap a state. The link is shareable: add `#Johor` to the address.

Haze right now



1. **Band** – Malaysian API bands: Good 0–50 · Moderate 51–100 · Unhealthy 101–200 · Very unhealthy 201–300 · Hazardous 300+. The colour follows the *worst* station.
2. **Four numbers** – highest API, national mean, stations above 100 / 150, and tomorrow's closure-risk count.
3. **Band scale** – the white marker is the national mean; the coloured marker is the worst station.
4. **State tile** – the state's highest reading. The more haze, the thicker the particles and the blurrier the skyline.

National outlook and schools impacted

MALAYSIA · OUTLOOK FOR TOMORROW MORNING

68/68 stations reporting · latest hour 08:00 MYT

1 Stations at closure risk
13 / 68
highest: Pasir Gudang, Johor (83%)

Schools in states at risk
3,238
5 of 16 states · 1,437,339 students · 128,209 teachers

2 Schools likely affected
2,616
closure chance × schools, state by state · 1,213,805 students

schools monitored
10,238
4,818,596 students · 417,462 teachers · 20 public universities

State	Status	Top station	Closure	7AM>150	Stns	# schools in area
Johor	Close / PdPR	Pasir Gudang	83%	90%	7 / 8	1,195
Negeri Sembilan	Close / PdPR	Nilai	79%	93%	2 / 3	479
Melaka	Close / PdPR	Bukit Rambai	78%	85%	1 / 3	318
Terengganu	Close / PdPR	Kemaman	73%	95%	1 / 4	506
Pahang	Close / PdPR	Rompin	48%	68%	2 / 5	740
WP Putrajaya	Watch	Putrajaya	12%	91%	0 / 1	27
Sarawak	Watch	Sri Aman	12%	77%	0 / 13	1,460
Selangor	Watch	Banting	9.8%	97%	0 / 6	945
WP Kuala Lumpur	Watch	Batu Muda	4.5%	92%	0 / 2	290
Sabah	Open	Tawau	0.1%	2.5%	0 / 6	1,298
Perak	Open	Seri Manjung	<0.1%	0.3%	0 / 5	1,105
Pulau Pinang	Open	Minden	<0.1%	0.2%	0 / 4	396
Kelantan	Open	Kota Bharu	<0.1%	0.1%	0 / 2	595
WP Labuan	Open	Labuan	<0.1%	0.1%	0 / 1	28
Kedah	Open	Sungai Petani	<0.1%	<0.1%	0 / 4	752
Perlis	Open	Kangar	<0.1%	<0.1%	0 / 1	104

3 **5** **4**

Click a state to open its stations. A state's closure chance is its highest-risk station; 'Stns' = stations at risk / reporting; '# schools in area' = government primary + secondary schools in the state; 'Schools likely affected' = closure chance × those schools, summed over states.

1. **Stations at closure risk** – stations whose closure probability is above the cost-optimal trigger.
2. **Schools likely affected** – closure chance × the number of schools in each state, added up (students and teachers shown underneath).
3. **Status** – Close / PdPR, Watch or Open per state, worst first. A state's chance is its highest-risk station.
4. **Click a row** to filter the page to that state and see its stations.
5. **# schools in area** – government primary + secondary schools in the state (state totals, MOE via OpenDOSM).

1

71%
closure risk

LARKIN · JOHOR · RECOMMENDATION

Close tomorrow · switch to PdPR

Closure risk 71% is above the cost-optimal trigger of 25%. Notify parents and hostels; re-check at 8 PM and 6 AM. ⚠ Inputs are outside the range seen in training – indicative only.

2

7 AM API > 100	7 AM API > 150	API > 180 in school hours	Streak rule
98%	**83%**	**7.4%**	**69%**
unhealthy	very unhealthy	closure trigger	tomorrow mean > 150

4

THREE SCENARIOS FOR TOMORROW MORNING

Best · improving	Base · unstable	Worst · closure
0.5%	**29%**	**71%**
7 AM reading ≤ 100. Announce open early; families can plan.	Unhealthy but below closure trigger. Open with restrictions; keep PdPR ready.	API > 180 in school hours, or 3rd straight day > 150. Close / PdPR.

1 **Recommendation** – open / open with restrictions / close-PdPR, with the closure ring and why.

2 **Probabilities** – 7 AM API above 100 and 150, API above 180 in school hours, and the 3-day streak rule (KPM's two closure triggers).

3 **Schools impacted (state level)** – schools, students and teachers in the state. "If schools close" = everyone affected by a state-wide closure; "Likely affected tomorrow" = closure chance × schools.

4 **Three scenarios** – best (improving), base (unstable), worst (closure). The framed one is the most likely.

STEP 6 · DATA CHECK **Is every DOE report backed by an independent source?**

DATA CHECK · MALAYSIA Is every DOE report backed by an independent source?

1

Stations flagged (last 5 days)	Closure-level readings unconfirmed	Possible under-reporting in haze	Reliability A · B · C
12 / 68	4	3	28 · 32 · 8
13 reports · 7 critical · 0 still active	DOE ≥ 150 that Open-Meteo does not support	Open-Meteo ≥ 150 while DOE < 100	median correlation 0.78 · 100.0% within one band

2

B	**Kuala Terengganu · Terengganu — Possible under-reporting** DOE 76 is below what Open-Meteo supports (≈156): possible under-reporting during haze	12 Oct 20:00 → 16:00 21 h · DOE 76 vs ≈156
C	**JOHAN SETIA · Selangor — Possible under-reporting** DOE 93 is below what Open-Meteo supports (≈164): possible under-reporting during haze	13 Oct 03:00 → 13:00 11 h · DOE 93 vs ≈164
B	**Sri Aman · Sarawak — Unconfirmed high reading** Unconfirmed closure-level reading: DOE 160 is above what Open-Meteo supports (≈46, up to 129)	12 Oct 15:00 → 02:00 12 h · DOE 160 vs ≈46
A	**Seremban · Negeri Sembilan — Unconfirmed high reading** Unconfirmed closure-level reading: DOE 163 is above what Open-Meteo supports (≈65, up to 134)	12 Oct 00:00 → 20:00 21 h · DOE 163 vs ≈65
A	**Nilai · Negeri Sembilan — Unconfirmed high reading** Unconfirmed closure-level reading: DOE 163 is above what Open-Meteo supports (≈67, up to 144)	12 Oct 06:00 → 17:00 12 h · DOE 163 vs ≈67
B	**Temerloh · Pahang — Possible under-reporting** DOE 72 is below what Open-Meteo supports (≈158): possible under-reporting during haze	11 Oct 02:00 → 08:00 7 h · DOE 72 vs ≈158
A	**Seberang Jaya · Pulau Pinang — Unconfirmed high reading** Unconfirmed closure-level reading: DOE 152 is above what Open-Meteo supports (≈74, up to 140)	9 Oct 09:00 → 15:00 7 h · DOE 152 vs ≈74
A	**Seberang Jaya · Pulau Pinang — Unconfirmed high reading** Unconfirmed high reading: DOE 149 is above what Open-Meteo supports (≈71, up to 134)	9 Oct 17:00 → 21:00 5 h · DOE 149 vs ≈71

Show all 13 flagged reports

3

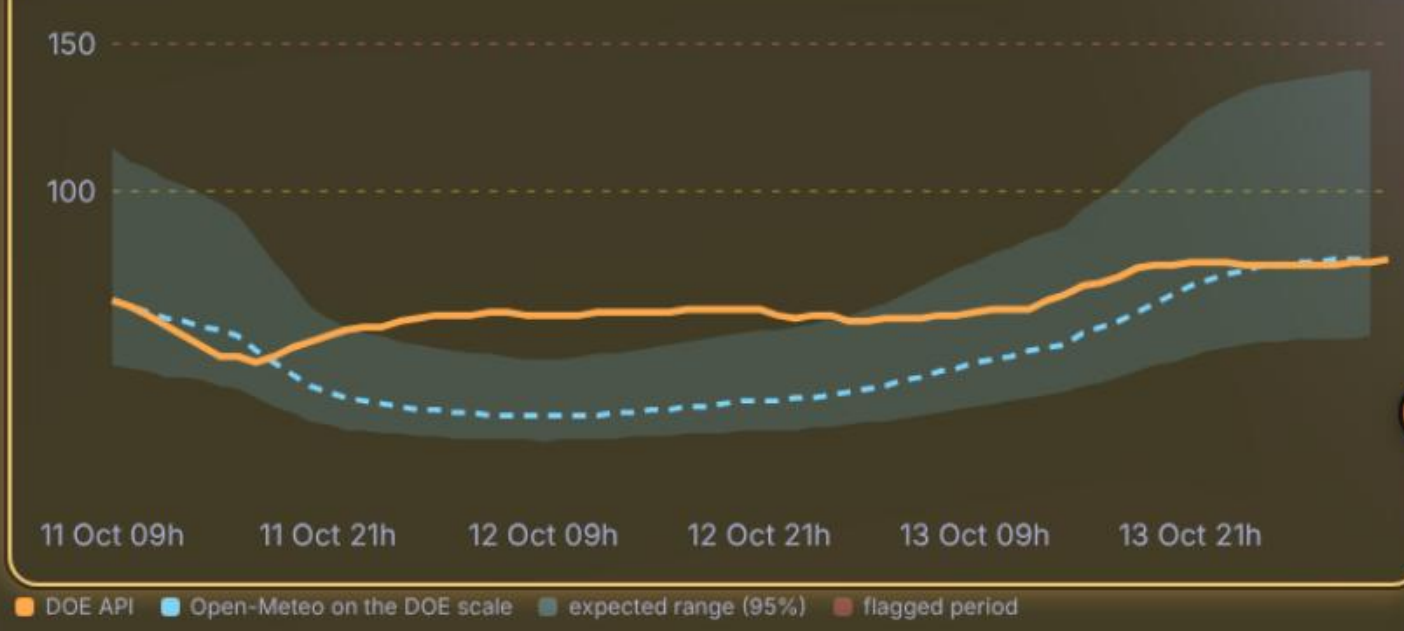
Next-day closure forecast, skill vs base rate	3 PM	4 PM	5 PM	6 PM	7 PM	8 PM	9 PM
From the DOE feed	66%	67%	67%	66%	68%	70%	72%
From Open-Meteo alone	51%	51%	52%	51%	52%	54%	54%

How to read this. Open-Meteo (Copernicus CAMS atmospheric model, ~25 km grid) is an independent estimate of the same air. Its 24-hour mean US AQI is mapped to each station's own DOE scale, and the range DOE normally falls in is measured from history. A flag means a *sustained, decision-relevant* gap (≥ 4 h, DOE at/over 100 or Open-Meteo at/over 100), a repeated value for a day, or a one-hour jump the other source does not share. **"Unconfirmed" is not "wrong"**: a model cannot see a single chimney or refinery, so local sources look unconfirmed. Grade C stations should be verified before closing schools. Contains modified Copernicus Atmosphere Monitoring Service information, via Open-Meteo.com (CC BY 4.0).

Open any flagged station: DOE against Open-Meteo on the same scale, plus a second-opinion closure chance:

INDEPENDENT CHECK · SERI MANJUNG, PERAK DOE vs Open-Meteo, last 72 h

4



5

Reliability vs Open-Meteo

A Agrees closely

correlation 0.757 · typical gap 9.6 pts · same band 81% · within one band 100.0%

High readings corroborated

22%

of 302 hours with DOE ≥ 150, Open-Meteo also ≥ 100

6

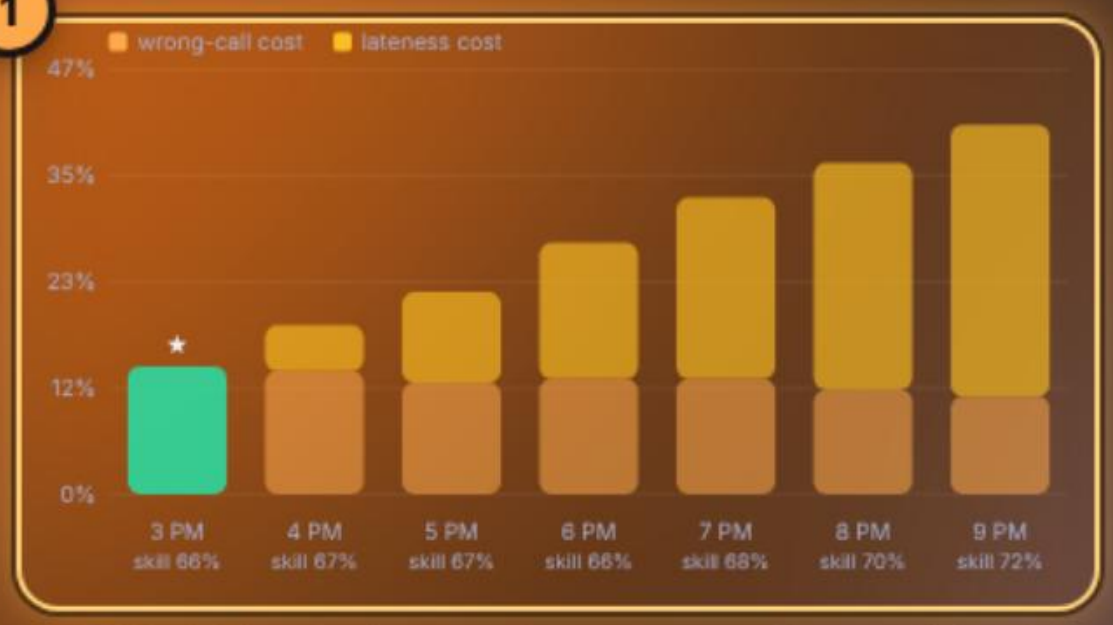
Closure chance · DOE feed	Closure chance · Open-Meteo only
<0.1%	<0.1%
trigger 25%	same side of the trigger

1 flagged report for this station in the last 5 days (shaded). DOE reported exactly 158 for 15 hours in a row while Open-Meteo moved 30 points (possible stuck sensor or repeated value).

1. **Summary** – stations flagged in the last 5 days, closure-level readings the independent source does not support, possible under-reporting during haze, and how many stations are reliability A / B / C.
2. **Flagged report** – what happened, where and when. ACTIVE = still going on. Click to open the station. "Unconfirmed" is not "wrong": a 25 km model cannot see a single local source.
3. **Open-Meteo alone as a forecaster** – skill of the next-day closure forecast built only from the independent source, next to the DOE feed. It is weaker, which is why it is a second opinion, not a replacement.
4. **Chart** – DOE (orange) against Open-Meteo mapped to the DOE scale (blue dashed) with the range DOE normally falls in. Red shading = flagged hours.
5. **Reliability grade** – A agrees closely · B usable · C verify before acting (large typical gap, poor band agreement, or high readings rarely corroborated).
6. **Second opinion** – closure chance from the DOE feed vs from Open-Meteo alone. Green frame = same side of the trigger; amber = they disagree, so verify.

The best announcement time reacts to your preferences

1 BEST ANNOUNCEMENT TIME



★ 3 PM minimises total expected cost at your preferences (missed closure = 8× needless closure; each hour later = 5% of baseline). Click a bar to explore that hour in What-if.

2 BACKTEST BY ANNOUNCEMENT TIME

Announce	Skill	AUC >100	AUC all	Err. cost
3 PM ★	66%	0.931	0.998	14%
4 PM	67%	0.935	0.998	14%
5 PM	67%	0.934	0.998	12%
6 PM	66%	0.940	0.998	13%
7 PM	68%	0.949	0.998	13%
8 PM	70%	0.951	0.998	12%
9 PM	72%	0.956	0.999	11%

Out-of-time hold-out 2025-07-01 → 2026-10-13: 30,025 station-days,

Now drag "Cost of each hour later" from 5% to 0% (delay is free) – the ★ moves from the earliest hour to the latest:

DECISION PREFERENCES

Wrongly staying open costs × a needless closure — 8

3 ost of each hour later of baseline — 0%

Higher cost of a missed closure moves the best time earlier; a higher cost of waiting does the

4 BEST ANNOUNCEMENT TIME



★ 9 PM minimises total expected cost at your preferences (missed closure = 8× needless closure; each hour later = 0% of baseline). Click a bar to explore that hour in What-if.

1. **Cost bars** – wrong-call cost (missed closure vs needless closure) plus lateness cost for each candidate hour; ★ = cheapest. Click a bar to explore that hour.
2. **Backtest table** – on days the model never saw: skill and ranking quality at each decision hour.
3. **Decision preferences** (left panel) – how bad a missed closure is, and what each hour of lateness costs.
4. **Result** – with free delay the ★ moves to 9 PM because later hours forecast better; with costly delay the earliest hour wins.

STEP 8 · TRY YOUR OWN NUMBERS

What-if mode

1 MODE

Live data | What-if

2 Model hour 11 PM

12 PM · 1 PM · 2 PM · 3 PM · 4 PM · 5 PM · 6 PM · 7 PM · 8 PM · 9 PM · 10 PM · 11 PM

What-if: edit any reading below.

3

PI at 8 AM 105

API at 12 PM 120

API at 8 PM 135

API at 11 PM 150

Yesterday's daily mean 172

4

ior consecutive days with mean > 150

0 · 1 · 2 · 3+

Editing Pasir Gudang, Johor.

5

5%
closure risk

PASIR GUDANG · JOHOR · RECOMMENDATION

Open with restrictions · PdPR on standby

Closure risk 4.6% is below the trigger (25%) but not negligible. Cancel outdoor activity and confirm at 8 PM / 6 AM. ⚠ Inputs are outside the range seen in training – indicative only.

7 AM API > 100	7 AM API > 150	API > 180 in school hours	Streak rule
100%	91%	4.6%	<0.1%
unhealthy	very unhealthy	closure trigger	needs ≥2 prior days > 150

THREE SCENARIOS FOR TOMORROW MORNING

Best · improving
0.3%
7 AM reading ≤ 100. Announce open early; families can plan.

Base · unstable
95%
Unhealthy but below closure trigger. Open with restrictions; keep PdPR ready.

Worst · closure
4.6%
API > 180 in school hours, or 3rd straight day > 150. Close / PdPR.

Best 0.3% · Base 95% · Worst 4.6%

TODAY'S TREND → TOMORROW 7 AM



SEASONALITY · JOHOR · SHARE OF STATION-DAYS WITH MEAN API > 100



1 **What-if** – switch here; sliders unlock. Switch back to Live data at any time.

2 **Model hour** – pick the hour you would decide at (outlined = candidate announcement times).

3 **Readings** – set API at 8 AM, noon, three hours earlier and now, plus yesterday's mean.

4 **Streak** – how many prior days had a mean above 150 (KPM's three-day rule).

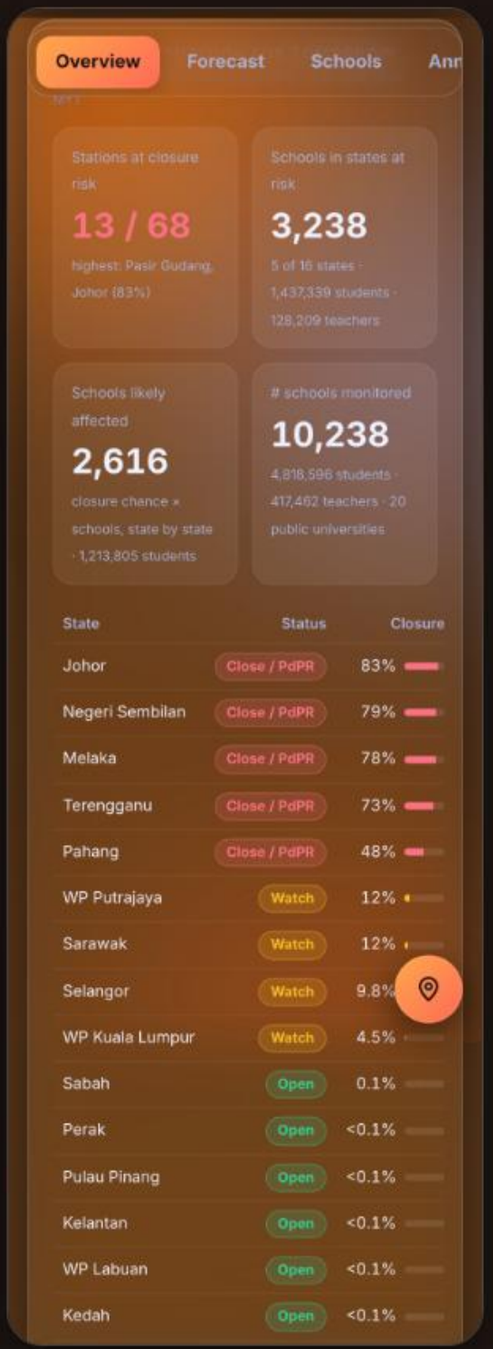
5 **Result** – the recommendation and every probability recompute instantly.

Illustrative inputs, not an official reading

Same dashboard, stacked for small screens



Haze right now



National outlook



Station recommendation